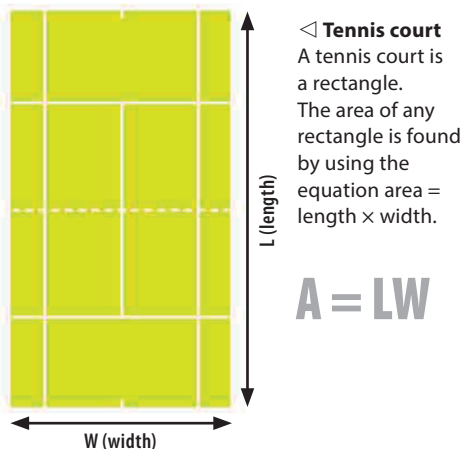


REAL WORLD

Algebra in everyday life

Although algebra may seem abstract, with equations consisting of strings of symbols and letters, it has many applications in everyday life. For example, an equation can be used to find out the area of something, such as a tennis court.

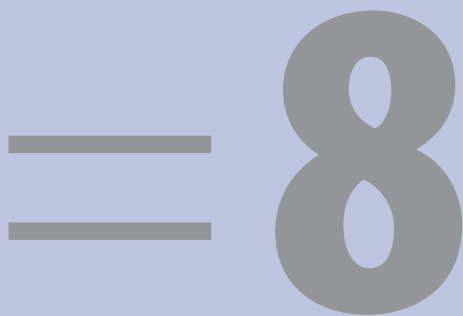


EQUALS

The equals sign means that the two sides of the equation balance each other

CONSTANT

A number with a value that is always the same



THE ANSWER IS:

$$b = 6$$

BASIC RULES OF ALGEBRA

Like other areas of maths, algebra has rules that must be followed to get the correct answer. For example, one rule is about the order in which operations must be done.

Addition and subtraction

Terms can be added together in any order in algebra. However, when subtracting, the order of the terms must be kept as it was given.

$$a + b = b + a$$

△ Two terms

When adding together two terms, it is possible to start with either term.

$$(a + b) + c = a + (b + c)$$

△ Three terms

As with adding two terms, three terms can be added together in any order.

Multiplication and division

Multiplying terms in algebra can be done in any order, but when dividing the terms must be kept in the order they were given.

$$a \times b = b \times a$$

△ Two terms

When multiplying together two terms, the terms can be in any order.

$$a \times (b \times c) = b \times (a \times c) = c \times (a \times b)$$

△ Three terms

Multiplication of three terms can be done in any order.